MATH317 · NUMERICAL METHODS

Midterm 462 Solutions

18/03/2025 · 60 minutes · 30 marks

Muharib

01 Solution Q1: errors and Horner

1a) Absolute error = $|\ 30.158 - 30.16\ | = 0.002$

1b) Relative error = $\dfrac{0.002}{30.158} = 0.66 \times 10^{-4}$

$= 0.0066\ \%$

2a)

	4	−8	14	−32
3 |		12	12	78
	4	4	26	46

$P(3) = 46$

2b)

	4	−8	14	−32	
2 |		8	0	28	
	4	0	14 |	−4	$c_0 = -4$
		8	16		
	4	8 |	30		$c_1 = 30$
		8			
	4 |	16			$c_2 = 16$
	4				$c_3 = 4$

$P(x) = 4(x-2)^3 + 16(x-2)^2 + 30(x-2) - 4$

02 Solution Q2: double precision

1. $(184)_{10} = (270)_8 = (1011\ 1000)_2$

2. $(.125)_{10} = (.1)_8 = (.001)_2$

3. $(184.125)_{10} = (1011\ 1000.001)_2 = (1.011\ 1000001)_2 \times 2^7$

   stored mantissa: .011 1000001

4. $c - 1023 = 7 \quad \rightarrow \quad c = 1030 = (2006)_8 = (100\ 0000\ 0110)_2$

5. $[1 \mid 10000000110 \mid 0111000001\ 000...0]_2$

   $[1100\ 0000\ 0110\ 0111\ 0000\ 0100\ 0000\ \ldots\ 0000]_2$

   $= [C067040000000000]_{16}$

Notes on the key

The key writes the answer as $[C0670400]_{16}$ without the trailing zeros. A full double-precision answer has 16 hex digits.

Step 3 of the key shows `1259.125` and step 4 shows `130`. Both are typos; the correct values are `184.125` and `1030`.

03 Solution Q3: Bisection

Part 1: number of iterations

$$n > \frac{\log(b - a) - \log(2\varepsilon)}{\log 2}$$

$$n > \frac{\log(1) - \log(2 \times 10^{-2})}{\log 2} = 5.64$$

The smallest integer that works is `n = 6`. Since counting starts at `n = 0`, that is `7` iterations. The key writes: `5.6 ≈ 7`.

Part 2: finding the root

$$u = f(0) = -1 \qquad v = f(2) = 8 - e^2 = 8 - 7.3891 = 0.6109$$

$$u \cdot v < 0 \quad \rightarrow \quad f \text{ has a root in } [0, 2]$$

$$n = 0: \quad c = \frac{0 + 2}{2} = 1 \qquad w = f(1) = -0.71828$$

$$u \cdot w > 0 \quad \rightarrow \quad a = c \quad \rightarrow \quad [1, 2]$$

$$n = 1: \quad c = \frac{1 + 2}{2} = 1.5 \qquad w = f(1.5) = 0.01831$$

$$u \cdot w < 0 \quad \rightarrow \quad b = c \quad \rightarrow \quad [1, 1.5]$$

$$n = 2: \quad c = \frac{1 + 1.5}{2} = 1.25 \qquad w = f(1.25) = -0.36534$$

$$u \cdot w > 0 \quad \rightarrow \quad a = c \quad \rightarrow \quad [1.25, 1.5]$$

n	c_n	$f(c_n)$	Error
0	1	−0.71828	1
1	1.5	0.01831	0.5
2	1.25	−0.36534	0.25

At `n = 1` the sign changes: `u` is now the negative `f(1)` and `w` is positive, so the root is in the left half.